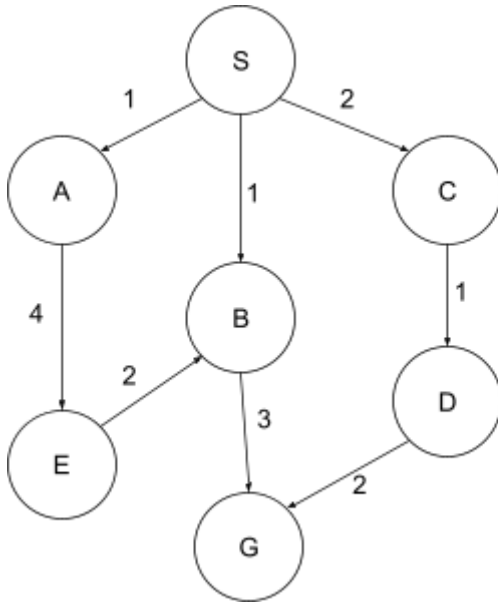


BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE422 : Artificial Intelligence
Assignment 01

- 1) a. For the state space tree given below, define the heuristic so that it is admissible.



States, n	$h(n)$
S	
A	
B	
C	
D	
E	
G	0

- b. Perform GBFS and A* Search using the heuristic values. **Show your steps clearly.**
- c. Using calculations where necessary, show whether the heuristic is also **consistent**. If not, update the values to be **consistent**.
- d. What problems of GBFS does A* Search solve?
- 2) Imagine you have a 5x5 gridworld, where an agent can move. You are running Graph A* Search from the **bottom-left** corner to the **top-right** corner. Horizontal and Vertical movements cost 1 and Diagonal movements cost $\sqrt{2}$. Heuristic $h_1(n)$ is the Manhattan Distance to the goal and $h_2(n)$ is the Euclidean Distance to the goal. Answer the following questions:
- Are the two heuristics admissible? Why or why not? Explain using any two nodes.
 - If $h_1(n)$ is multiplied by 1.2, will it still be admissible? Will A* Search still be optimal?
 - When we have multiple types of heuristics for a single problem, which one do we take and why?